

Savitribai Phule Pune University		
Second Year of Computer Science and Engineering - 2024 Pattern		
PCC-253-CSE : Computer Organization and Microprocessor		
Teaching Scheme	Credits	Examination Scheme
Theory : 02 Hours/Week	02	CCE : 30 Marks End-Semester: 70 Marks

Prerequisite Courses : Digital Electronics and Logic Design

Course Objectives: To provide students with a foundational understanding of computer evolution, memory management, the 8086 microprocessor, memory organization, interrupts, and parallel organization in computer systems.

1. Learn about the history and development of computers and how their performance has improved over time.
2. Understand how memory is managed in a computer and the techniques used to store and access data efficiently.
3. Get familiar with the 8086 microprocessor, its basic functions, and its role in computer systems.
4. Learn how memory is organized in a computer and how interrupts help manage tasks.
5. Understand the basics of parallel organization and how it improves the speed and performance of computer systems.

Course Outcomes: Upon successful completion of this course, students will be able to:

- CO1: Understand and analyze computer system design and historical development of computers and foundational architectures like Von Neumann and Harvard.
- CO2: To design and optimize internal, and external memory systems for high-performance computing.
- CO3: To understand fundamental knowledge of the 8086-microprocessor architecture, programming model, addressing modes, and instruction set.
- CO4: To understand memory management, I/O operations and interrupt handling, including address translation, memory banks and interrupt controllers.
- CO5: To explore the multiprocessor systems, Flynn's Taxonomy and RISC design principles, including memory architectures

Course Contents

Unit I - Computer Evolution and Performance (06 Hours)

A Brief History of Computers, Von Neumann Architecture, Harvard Architecture, Designing for Performance, Evolution of Intel processor architecture- 4 bit to 64 bit, performance assessment. A top level view of Computer function and interconnection Computer Components, Interconnection structure, bus interconnection,

Computer Arithmetic- The Arithmetic and Logic Unit, addition and subtraction of signed numbers, design of adder and fast adder, carry look ahead addition, multiplication of positive numbers, signed operand multiplication, Booths algorithm for multiplying binary integers.

Case Study: Evolution (Brief History) of Microprocessors

Unit II - Memory Management (06 Hours)

Characteristics of Memory System, The memory hierarchy.

Cache Memory- Cache memory principles, Elements of cache design cache address, size, and mapping functions. Replacement algorithms, write policy, line size, number of cache, one level and two level cache. Performance characteristics of two level cache- locality & operations.

Internal Memory- semiconductor main memory, advanced DRAM organization. External Memory- Hard Disk organization. RAID- level 1 to level 6.

Case Study : Memory Management in ATMs

Unit III - Introduction to 8086 Microprocessor (06 Hours)

8086 Architecture: Introduction to 16 bit microprocessor, Architecture and Pin diagram of 8086, Programmers model of 8086 (Registers).

Addressing modes of 8086: Immediate Addressing, Register Addressing, Direct Addressing, Indirect Addressing, Indexed Addressing, Based Addressing, Based Indexed Addressing

Instruction set of 8086: Data Movement Instructions, Arithmetic Instructions, Logic Instructions, Control Transfer Instructions, String Instructions, Input / Output Instructions, Flag Control Instructions, Process Control Instructions, Other Instructions.

Case study: Design of Basic Calculator Using 8086

Unit IV - Memory Organization and Interrupts (06 Hours)

Memory Organization: Segmentation, logical to physical address translation, even and odd memory banks, Read write cycle timing diagrams, Address mapping and decoding, I/O: memory mapped I/O & I/O Mapped I/O.

Interrupts: Interrupt Control & status registers, Interrupt Vector Table (IVT), ISR, Hardware and software Interrupts, 8259 (Programmable Interrupt Controller): Features, Block Diagram, Control & Status registers.

Case study : Memory and Interrupt Management in an Automated Railway Reservation System

Unit V - Parallel Organization (06 Hours)

Multiprocessors, Clusters, Flynn's Taxonomy for Multiple Processor Organizations, Closely and Loosely Coupled Multiprocessors Systems, Symmetric Multiprocessor (SMP) Organization, UMA, NUMA.

RISC: Instruction execution characteristics, use of large register file, compiler-based register optimization, RISC architecture and pipelining. RISC Vs CISC.

Case study: Multi-core System

Learning Resources

Text Books:

1. W. Stallings, Computer Organization and Architecture: Designing for performance, Pearson Education/ Prentice Hall of India, 2003, ISBN 978-93-325-1870-4, 7 th Edition.
2. Zaky S, Hamacher, —Computer Organization, 5th Edition, McGraw-Hill Publications, 2001, ISBN- 978-1-25-900537-5, 5th Edition.
3. Douglas Hall, "Microprocessors & Interfacing", McGraw Hill, Revised 2 Edition,
4. A. Ray, K. Bhurchandi, "Advanced Microprocessors and peripherals: Arch, Programming & Interfacing" Tata McGraw Hill, 2004 ISBN 0-07-463841.

Reference Books:

1. John P Hays, —Computer Architecture and Organization, McGraw-Hill Publication, 1998, ISBN:978-1-25-902856-4, 3rd Edition.
2. Miles Murdocca and Vincent Heuring, —Computer Architecture and Organization- an integrated approach, Wiley India Pvt. Ltd, ISBN:978-81-265-1198-3, 2nd Edition

3. A. Tanenbaum, —Structured Computer Organization||, Prentice Hall of India, 1991 ISBN: 81 – 203 – 1553 – 7, 4th Edition.
4. Patterson and Hennessy, —Computer Organization and Design, Morgan Kaufmann Publishers In, ISBN 978-0-12-374750-1, 4th Edition.
5. Liu, Gibson, “Microcomputer Systems: The 8086/88 Family”, 2 nd Edition, PHI,2005.
6. Kenneth Ayala, “The 8086 Microprocessor: Programming & Interfacing the PC”

MOOC / NPTEL/YouTube Links: -

1. NPTEL course on “Computer architecture and organization” By Prof. Indranil Sengupta, Prof. Kamalika Datta, IIT Kharagpur https://swayam.gov.in/nd1_noc21_cs61/preview
2. NPTEL course on “Microprocessor & Interfacing” By Prof. Shaik Rafi Ahamed, IIT Guwahati <https://archive.nptel.ac.in/courses/108/103/108103157/>
3. Complete COA Computer Organization & Architecture in one shot, Semester Exam, Hindi, <https://www.youtube.com/watch?v=DsK35f8wyUw>
4. 8086 Microprocessor Architecture - Bharat Acharya, Hindi, <https://www.youtube.com/watch?v=Dmw>
5. Microprocessor 8086 - YouTube Playlist https://www.youtube.com/playlist?list=PLgwJf8NK-2e4oAeDid0hwuiol_RJdscrp